Logic — Class Notes

November 6, 2025

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1 Generating Functions

Definition

A **generating function** is a formal power series in one variable (usually x) whose coefficients correspond to the terms of a sequence.

If $\{a_n\}$ is a sequence, then its generating function is defined as

$$G(x) = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \cdots = \sum_{n=0}^{\infty} a_n x^n.$$

Generating functions are useful tools to study sequences, solve recurrence relations, and count combinatorial objects.

Example 1. Constant sequence

Let $a_n = 1$ for all $n \geq 0$. Then

$$G(x) = 1 + x + x^2 + x^3 + \cdots = \frac{1}{1-x}, \quad |x| < 1.$$

This is the generating function for the sequence $\{1, 1, 1, 1, \dots\}$.

Example 2. Arithmetic sequence

Let $a_n = n$ for $n \geq 0$. Then

$$G(x) = 0 + 1x + 2x^2 + 3x^3 + \cdots$$

Example 3. Geometric sequence

Let $a_n = r^n$ for some constant r. Then

$$G(x) = 1 + rx + r^2x^2 + r^3x^3 + \cdots$$

Hence, the generating function represents a geometric progression.

Example 4. Sequence of even numbers

Let $a_n = 2n$ for $n \geq 0$. Then

$$G(x) = 0 + 2x + 4x^2 + 6x^3 + \cdots$$

Remarks:

- Generating functions turn sequence problems into algebraic problems.
- They help find closed forms, solve recurrences, and count combinations.
- There are many types: ordinary generating functions (OGFs), exponential generating functions (EGFs), and others.

2 Inclusion–Exclusion Principle

Formulas (Inclusion–Exclusion)

- **Two sets:** For finite sets A, B,

$$|A \cup B| = |A| + |B| - |A \cap B|.$$

- **Three sets:** For finite sets A, B, C,

$$|A \cup B \cup C| = |A| + |B| + |C| - \big(|A \cap B| + |A \cap C| + |B \cap C|\big) + |A \cap B \cap C|.$$

- **Exactly one of three sets:**

$$|\,\text{exactly one of } A, B, C\,| = |A|+|B|+|C|-2\big(|A\cap B|+|A\cap C|+|B\cap C|\big)+3|A\cap B\cap C|.$$

- **Exactly two of three sets:**

$$|\,\text{exactly two}\,| = |A \cap B| + |A \cap C| + |B \cap C| - 3|A \cap B \cap C|.$$

1. In a class of 50 students, 28 study Mathematics and 30 study Physics. If 20 study both Mathematics and Physics, how many students study neither subject?

   Solution. Let M = students who study Mathematics, P = students who study Physics. Use
   $$|M \cup P| = |M| + |P| - |M \cap P| = 28 + 30 - 20 = 38.$$
   Total is 50, so number who study neither $= 50 - 38 = 12$.

2. Out of 80 people, 45 like coffee, 50 like tea, and 20 like both. How many like exactly one of the two beverages?

   Solution. Exactly one $= |Coffee| + |Tea| - 2|Coffee \cap Tea|$ because those who like both are counted twice if we add singles. Thus
   $$\text{exactly one} = 45 + 50 - 2 \cdot 20 = 55.$$
   (Alternatively: exactly one $= |Coffee \cup Tea| - |Coffee \cap Tea| = (45+50-20)-20 = 55$.)

3. In a college, 40 students play football, 30 play basketball, 20 play cricket. Also 10 play both football and basketball, 8 play both basketball and cricket, 5 play both football and cricket, and 3 play all three. How many students play at least one sport?

   Solution. Let F, B, C denote the three sets. Apply the three-set formula:
   $$\begin{aligned} |F \cup B \cup C| &= 40 + 30 + 20 - (10 + 8 + 5) + 3 \\ &= 90 - 23 + 3 = 70. \end{aligned}$$
   So 70 students play at least one sport.

4. In a survey, 60% like product A, 50% like product B, and 30% like both. What percent like at least one product?

   Solution. Using two-set formula on percentages,
   $$P(A \cup B) = 60 + 50 - 30 = 80\%.$$

5. Among 200 people, 120 own a car, 100 own a bike, 70 own a bicycle. Also 40 own both car and bike, 30 own both bike and bicycle, 20 own both car and bicycle, and 10 own all three. (a) How many own at least one vehicle? (b) How many own exactly one vehicle?

   Solution. Let C, Bi, By denote car, bike, bicycle.

   $$|C \cup Bi \cup By| = 120 + 100 + 70 - (40 + 30 + 20) + 10 = 290 - 90 + 10 = 210.$$

   Note: result 210 exceeds total population 200 → this indicates data inconsistency (over-counting beyond population) or survey respondents could select multiple options — but since population is 200, the maximum possible at least-one is 200. In real problems ensure totals are consistent. If we proceed formally ignoring population cap, at least-one = 210.

   (If the intent was to compute exact-one assuming the numbers are consistent:)

   Exactly one:

   $$\begin{aligned}\text{exactly one} &= |C| + |Bi| + |By| - 2(|C \cap Bi| + |Bi \cap By| + |C \cap By|) + 3|C \cap Bi \cap By| \\ &= 120 + 100 + 70 - 2(40 + 30 + 20) + 3 \cdot 10 \\ &= 290 - 180 + 30 = 140.\end{aligned}$$

   (Interpretation note: if universe is 200, exactly-one cannot exceed 200; check data consistency.)

6. In a company of 500 employees, 300 use Gmail, 250 use Outlook, and 100 use both. How many use neither service?

   Solution. Let G and O be the two sets. First compute union:

   $$|G \cup O| = 300 + 250 - 100 = 450.$$

   Employees using neither $= 500 - 450 = 50$.

7. In a class of 90 students, 40 take French, 50 take Spanish, 20 take both French and Spanish. Additionally, 15 take both French and German, 10 take both Spanish and German, and 5 take all three. How many take exactly two languages?

   Solution. Use the formula for exactly two:

   $$\begin{aligned}\text{exactly two} &= |F \cap S| + |F \cap G| + |S \cap G| - 3|F \cap S \cap G| \\ &= 20 + 15 + 10 - 3 \cdot 5 = 45 - 15 = 30.\end{aligned}$$

   So 30 students take exactly two languages.

8. How many integers from 1 to 100 (inclusive) are divisible by 2 or 3 or 5?

   Solution. Let A = multiples of 2, B = multiples of 3, C = multiples of 5 in $\{1, \ldots, 100\}$.

   Compute sizes:

   $$|A| = \left\lfloor \frac{100}{2} \right\rfloor = 50, \quad |B| = \left\lfloor \frac{100}{3} \right\rfloor = 33, \quad |C| = \left\lfloor \frac{100}{5} \right\rfloor = 20.$$

   Pairwise intersections (multiples of lcm):

   $$|A \cap B| = \left\lfloor \frac{100}{6} \right\rfloor = 16, \quad |A \cap C| = \left\lfloor \frac{100}{10} \right\rfloor = 10, \quad |B \cap C| = \left\lfloor \frac{100}{15} \right\rfloor = 6.$$

Triple intersection (multiples of lcm(2,3,5)=30):

$$|A \cap B \cap C| = \left\lfloor \frac{100}{30} \right\rfloor = 3.$$

Now inclusion–exclusion:

$$\begin{aligned} |A \cup B \cup C| &= 50 + 33 + 20 - (16 + 10 + 6) + 3 \\ &= 103 - 32 + 3 = 74. \end{aligned}$$

So there are 74 integers between 1 and 100 divisible by 2, 3, or 5.

Notes for students:

- Always identify sets clearly and compute single set sizes, pairwise intersections, and triple intersections before applying the formula.
- For "exactly one" or "exactly two" problems, use the derived formulas above (or compute via union and subtract appropriate overlaps).
- Watch for inconsistent data: counts that imply more people than exist indicate either an interpretational issue or a mistake in the given numbers; discuss such inconsistencies in your solution.

3 Introduction to Logic

Definition

Logic is the formal study of reasoning. In mathematics and computer science, logic provides precise language and rules to represent and manipulate statements that are either true or false.

Reasoning is the process of drawing conclusions from premises or evidence. It can be classified as **informal (natural language) reasoning** or **formal logical reasoning**, depending on the language and structure used.

Informal (Natural Language) Reasoning

This type of reasoning occurs in everyday communication, where arguments are expressed in ordinary language. It often depends on context, tone, and shared understanding. While it is flexible and intuitive, it may suffer from **ambiguity**, **vagueness**, or **logical gaps** due to the imprecise nature of natural language.

Example

If it rains, the ground will be wet. It looks cloudy, so the ground will probably be wet.
This reasoning is plausible but not strictly logical; the conclusion depends on interpretation and assumption.

Formal Logical Reasoning

In formal reasoning, statements are expressed using precise **symbols and rules of logic**. Each statement has a clear truth value (true or false), and arguments are evaluated using structured methods such as truth tables or inference rules.

Formal logic eliminates ambiguity by using well-defined symbols such as $\wedge$ (and), $\vee$ (or), $\neg$ (not), and $\rightarrow$ (implies).

Example

Let p: "It rains."
Let q: "The ground is wet."
Then the statement "If it rains, the ground will be wet" can be expressed as $p \rightarrow q$.
This form allows the argument to be tested and validated systematically using rules of logic.

In summary: Informal reasoning emphasizes *understanding and communication*, while formal reasoning emphasizes *clarity, precision, and the avoidance of ambiguity*. Formal logic provides the framework necessary to ensure that conclusions follow necessarily from premises.

4 Propositions and Logical Statements

Definition

A *proposition* (or logical statement) is a declarative sentence that is either true or false, but not both.

Examples: "2+2=4" is (true)
"There exists a prime even number greater than 2" is (false).

Counter-examples: questions, commands, sentences with vague truth-value.

5 Logical Connectives and Truth Tables

We build complex propositions using connectives.

- Negation: $\neg p$ (read: not p)
- Conjunction: $p \wedge q$ (read: p and q)
- Disjunction: $p \vee q$ (inclusive or)
- Exclusive OR: $p \oplus q$ (true when exactly one of p, q true)
- Implication: $p \rightarrow q$ (if p then q)
- Biconditional: $p \leftrightarrow q$ (if and only if)

Example

Truth table for common connectives :

p	q	$\neg p$	$p \wedge q$	$p \vee q$	$p \rightarrow q$	$p \leftrightarrow q$
T	T	F	T	T	T	T
T	F	F	F	T	F	F
F	T	T	F	T	T	F
F	F	T	F	F	T	T

6 Tautology, Contradiction, Contingency

Definition

A formula is a *tautology* if it is true for all truth assignments.
A *contradiction* is false for all assignments.
Otherwise the formula is *contingent*.

Example: $p \vee \neg p$ is a tautology. $p \wedge \neg p$ is a contradiction.

7 Logical Equivalence and Laws of Logic

Two formulas A and B are *logically equivalent* (written $A \equiv B$) if $A \leftrightarrow B$ is a tautology.

Important equivalences :

- Commutative laws: $p \wedge q \equiv q \wedge p$, $p \vee q \equiv q \vee p$.
- Associative laws: $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$, similarly for $\vee$.
- Distributive laws: $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$.
- De Morgan's laws: $\neg(p \wedge q) \equiv \neg p \vee \neg q$, $\neg(p \vee q) \equiv \neg p \wedge \neg q$.
- Implication rewrite: $p \to q \equiv \neg p \vee q$.

Verification of Fundamental Logical Laws Using Truth Tables

1. Commutative Laws

$$p \wedge q \equiv q \wedge p \qquad \text{and} \qquad p \vee q \equiv q \vee p$$

p	q	$p \wedge q$	$q \wedge p$
T	T	T	T
T	F	F	F
F	T	F	F
F	F	F	F

p	q	$p \vee q$	$q \vee p$
T	T	T	T
T	F	T	T
F	T	T	T
F	F	F	F

Both columns of each table are identical, hence the laws hold true.

2. Associative Laws

$$(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$$

p	q	r	$(p \wedge q) \wedge r$	$p \wedge (q \wedge r)$
T	T	T	T	T
T	T	F	F	F
T	F	T	F	F
F	T	T	F	F

Both sides produce identical truth values for all cases. Similarly for $\vee$.

3. Distributive Law

$$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$$

p	q	r	$p \wedge (q \vee r)$	$(p \wedge q) \vee (p \wedge r)$
T	T	T	T	T
T	T	F	T	T
T	F	T	T	T
T	F	F	F	F
F	T	T	F	F

Thus, both expressions are equivalent.

4. De Morgan's Laws

$$\neg(p \wedge q) \equiv \neg p \vee \neg q \qquad \text{and} \qquad \neg(p \vee q) \equiv \neg p \wedge \neg q$$

p	q	$\neg(p \wedge q)$	$\neg p \vee \neg q$
T	T	F	F
T	F	T	T
F	T	T	T
F	F	T	T

p	q	$\neg(p \vee q)$	$\neg p \wedge \neg q$
T	T	F	F
T	F	F	F
F	T	F	F
F	F	T	T

5. Implication Rewrite

$$p \to q \equiv \neg p \vee q$$

p	q	$p \to q$	$\neg p \vee q$
T	T	T	T
T	F	F	F
F	T	T	T
F	F	T	T

Hence, $p \to q \equiv \neg p \vee q$ is verified.

Example

Simplify: $\neg(p \to q)$. Using implication rewrite: $\neg(\neg p \vee q) \equiv p \wedge \neg q$.

$$\neg(p \to q)$$

Using the equivalence $p \to q \equiv \neg p \vee q$, we have

$$\neg(p \to q) \equiv \neg(\neg p \vee q)$$

Applying De Morgan's law:

$$\neg(\neg p \vee q) \equiv p \wedge \neg q$$

$$\therefore \ \neg(p \to q) \equiv p \wedge \neg q$$

8 Arguments and Validity

Definition

An *argument* is a sequence of propositions: premises $P_1, \ldots, P_n$ and a conclusion C. The argument is *valid* if whenever all premises are true, the conclusion must be true.

Formal Test for Validity

To test whether an argument is **valid**, we check if the conclusion necessarily follows from the premises.

An argument with premises $P_1, P_2, \ldots, P_n$ and a conclusion C is said to be valid **if and only if**

$$(P_1 \land P_2 \land \cdots \land P_n) \to C$$

is a **tautology**, i.e., it is true for every possible combination of truth values.

In Simple Words

If all the premises are true, the conclusion must also be true. If there is even one case where all the premises are true but the conclusion is false, the argument is **invalid**.

Thus, the formal test for validity involves:

1. Combining all premises using the logical AND ($\land$).
2. Forming an implication to the conclusion ($\to$).
3. Constructing a truth table.
4. If the final column is always true, the argument is **valid**.

Example: Formal Test for Validity

Argument:

1. If it rains, the ground is wet.
2. It is raining.
3. Therefore, the ground is wet.

Let p: "It rains" and q: "The ground is wet."

The argument can be written as:

$$(p \to q),\ p\ \therefore q$$

To test validity, form the implication:

$$((p \to q) \land p) \to q$$

Construct the truth table:

p	q	$p \to q$	$(p \to q) \land p$	$((p \to q) \land p) \to q$
T	T	T	T	**T**
T	F	F	F	**T**
F	T	T	F	**T**
F	F	T	F	**T**

The final column is true in all rows. Therefore,

$$((p \to q) \land p) \to q$$

is a **tautology**, and the argument is **valid**.

Example: Invalid Argument

Argument:

1. If it rains, the ground is wet.
2. The ground is wet.
3. Therefore, it is raining.

Let p: "It rains" and q: "The ground is wet."

The argument can be expressed as:

$$(p \to q),\ q\ \therefore p$$

To test validity, form the implication:

$$((p \to q) \land q) \to p$$

Construct the truth table:

p	q	$p \to q$	$(p \to q) \land q$	$((p \to q) \land q) \to p$
T	T	T	T	**T**
T	F	F	F	**T**
F	T	T	T	**F**
F	F	T	F	**T**

Since the final column contains a **False** value (when $p = F, q = T$), the statement

$$((p \to q) \land q) \to p$$

is **not a tautology**, and therefore the argument is **invalid**.

9 Predicates and Quantifiers

Predicates and Quantifiers

1. Predicate

A **predicate** is a statement involving variables that becomes a proposition when specific values are substituted. It expresses a property or relation involving one or more objects.

$$P(x) : \text{“}x > 5\text{”}$$

If $x = 7$, then $P(7)$ is true. If $x = 3$, then $P(3)$ is false.

2. Domain of Discourse

The **domain** of a predicate is the set of values the variable can take.

$$\text{Domain} = \mathbb{Z}, \quad P(x) : x > 5$$

3. Quantifiers

- **Universal quantifier** ($\forall$): means "for all".

$$\forall x \in \mathbb{R},\ x^2 \geq 0$$

- **Existential quantifier** ($\exists$): means "there exists".

$$\exists x \in \mathbb{R},\ x^2 = 9$$

4. Negation Rules

$$\neg(\forall x\, P(x)) \equiv \exists x\, (\neg P(x)), \quad \neg(\exists x\, P(x)) \equiv \forall x\, (\neg P(x))$$

5. Examples

$$\forall x, x < x + 1 \quad \text{True}$$
$$\exists x, x^2 = 2 \quad \text{False (for integers)}$$

6. Nested Quantifiers

$$\forall x \exists y, (x < y) \quad \text{True in } \mathbb{R}$$
$$\exists y \forall x, (x < y) \quad \text{False in } \mathbb{R}$$

Statement 1:

$$\forall x\, \exists y\, (x < y)$$

Explanation: For every real number x, we can find some y greater than x. For example, choose $y = x + 1$. Hence, the statement is true because the set of real numbers has no largest element.

Statement 2:

$$\exists y\, \forall x\, (x < y)$$

Explanation: This statement says: "There exists a real number y that is greater than every real number x." But this would imply that $\mathbb{R}$ has a largest element, which is not true — the real numbers are unbounded above. For any proposed y, we can always find $x = y + 1$, which is greater than y, contradicting $x < y$. Hence, no such y exists, and the statement is false.

Summary

Quantifier	**Symbol**	**Read as**	**Meaning**
Universal	$\forall x\, P(x)$	For all x	$P(x)$ is true for all x
Existential	$\exists x\, P(x)$	There exists x	$P(x)$ true for some x
Negation of $\forall$	$\neg(\forall x\, P(x))$		$\exists x\, \neg P(x)$
Negation of $\exists$	$\neg(\exists x\, P(x))$		$\forall x\, \neg P(x)$

Theorem Proving Methods

When we want to prove that a conclusion C follows from premises $P_1, P_2, \ldots, P_n$, we aim to show that

$$(P_1 \wedge P_2 \wedge \cdots \wedge P_n) \implies C$$

is a tautology.

1. Direct Method (Rule of Inference)

Assume all premises are true and apply valid inference rules (Modus Ponens, Modus Tollens, etc.) step by step until reaching the conclusion.

Example 1

If it rains, the ground is wet. It rains. Therefore, the ground is wet.
$R \to W, R \Rightarrow W$

Example 2

If a number is divisible by 4, it is even. 12 is divisible by 4. Therefore, 12 is even.
$D(4, x) \to E(x), D(4, 12) \Rightarrow E(12)$

2. Indirect Method (Contrapositive)

To prove $p \to q$, prove its contrapositive $\neg q \to \neg p$.

Example 1

If a number is even, its square is even.
Contrapositive: If the square is odd, the number is odd.

Example 2

If an animal is a dog, it is a mammal.
Contrapositive: If not a mammal, then not a dog.

3. Proof by Contradiction

Assume the negation of the statement and show it leads to a contradiction.

Example 1

Prove $\sqrt{2}$ is irrational. Assume it is rational; derive contradiction since both a, b even.

Example 2

Assume a smallest positive rational number r exists. Then $r/2$ is smaller – contradiction.

4. Equivalence Method

To prove $p \equiv q$, show both $p \to q$ and $q \to p$.

Example 1

$\neg(p \vee q) \equiv (\neg p \wedge \neg q)$ — by De Morgan's Law.

Example 2

$p \leftrightarrow q \equiv (p \to q) \wedge (q \to p)$

10 Additional Examples on Theorem Proving Methods

1. Direct Proof

Example 1: Prove that if n is even, then n^2 is even.

Proof: Let $n = 2k$ for some integer k. Then, $n^2 = (2k)^2 = 4k^2 = 2(2k^2)$, which is even. Hence, if n is even, then n^2 is even. $\square$

Example 2: Prove that the sum of two odd integers is even.

Proof: Let the two odd integers be $a = 2m+1$ and $b = 2n+1$. Then, $a+b = 2m+1+2n+1 = 2(m+n+1)$, which is even. $\square$

Example 3: Prove that if x and y are rational, then $x + y$ is rational.

Proof: Let $x = \frac{p}{q}$ and $y = \frac{r}{s}$, where p, q, r, s are integers and $q, s \neq 0$. Then, $x + y = \frac{ps+rq}{qs}$, which is rational. $\square$

2. Proof by Contrapositive

Example 1: Prove that if n^2 is even, then n is even.

Proof: Contrapositive: If n is odd, then n^2 is odd. Let $n = 2k + 1$. Then, $n^2 = (2k + 1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1$, which is odd. Hence, the contrapositive is true, and so is the original statement. $\square$

Example 2: Prove that if $3n + 2$ is odd, then n is odd.

Proof: Contrapositive: If n is even, then $3n+2$ is even. Let $n = 2k$, then $3n+2 = 3(2k)+2 = 6k + 2 = 2(3k + 1)$, which is even. $\square$

Example 3: Prove that if x^2 is divisible by 3, then x is divisible by 3.

Proof: Contrapositive: If x is not divisible by 3, then x^2 is not divisible by 3. Let $x = 3k + 1$ or $x = 3k+2$. Then, $x^2 = 9k^2+6k+1$ or $9k^2+12k+4$, both leave a remainder 1 when divided by 3. Hence, x^2 is not divisible by 3. $\square$

3. Proof by Contradiction

Example 1: Prove that $\sqrt{2}$ is irrational.

Proof: Assume the contrary: $\sqrt{2} = \frac{p}{q}$, where p and q have no common factors. Then, $2 = \frac{p^2}{q^2}$, so $p^2 = 2q^2$. Hence, p^2 is even $\Rightarrow p$ is even $\Rightarrow p = 2r$. Substitute: $(2r)^2 = 2q^2 \Rightarrow q^2 = 2r^2$, so q is even. Both p and q are even — contradiction. Hence, $\sqrt{2}$ is irrational. $\square$

Example 2: Prove that there is no smallest positive real number.

Proof: Assume there is a smallest positive real number r. Then, $\frac{r}{2}$ is positive and smaller than r. Contradiction. Hence, no smallest positive real number exists. $\square$

Example 3: Prove that there is no integer between 2 and 3.

Proof: Assume there exists an integer n such that $2 < n < 3$. No such integer exists in $\mathbb{Z}$. Contradiction. $\square$

4. Biconditional Proof

Example 1: Prove that n is even if and only if n^2 is even.

Proof: ($\Rightarrow$) If n is even, $n = 2k$, then $n^2 = 4k^2 = 2(2k^2)$, which is even. ($\Leftarrow$) If n^2 is even, then by the contrapositive of Example 1 above, n is even. $\square$

Example 2: Prove that n is divisible by 6 if and only if n is divisible by both 2 and 3.

Proof: ($\Rightarrow$) If n is divisible by 6, then $n = 6k = 2(3k)$, so n is divisible by 2 and 3. ($\Leftarrow$) If n is divisible by both 2 and 3, say $n = 2a = 3b$, then $n = 6k$ for some k. $\square$

Example 3: Prove that a triangle is equilateral if and only if it is equiangular.

Proof: ($\Rightarrow$) If a triangle is equilateral, all sides are equal $\Rightarrow$ all angles equal (by congruence). ($\Leftarrow$) If all angles are equal, all sides are equal (by converse of the same theorem). $\square$

11 Worked Examples and Problems

(A) Equivalences

Example 1. Prove that $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$ using a truth table.

p	q	r	$q \vee r$	$p \wedge (q \vee r)$	$(p \wedge q) \vee (p \wedge r)$	Same?
T	T	T	T	T	T	T
T	T	F	T	T	T	T
T	F	T	T	T	T	T
T	F	F	F	F	F	T
F	T	T	T	F	F	T
F	T	F	T	F	F	T
F	F	T	T	F	F	T
F	F	F	F	F	F	T

Hence, $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$.

Example 2. Prove $\neg(p \vee q) \equiv \neg p \wedge \neg q$ (De Morgan's Law).

p	q	$p \vee q$	$\neg(p \vee q)$	$\neg p$	$\neg q$	$\neg p \wedge \neg q$
T	T	T	F	F	F	F
T	F	T	F	F	T	F
F	T	T	F	T	F	F
F	F	F	T	T	T	T

Thus, both columns are identical, proving $\neg(p \vee q) \equiv \neg p \wedge \neg q$.

Example 3. Prove $\neg(p \wedge q) \equiv \neg p \vee \neg q$.

p	q	$p \wedge q$	$\neg(p \wedge q)$	$\neg p$	$\neg q$	$\neg p \vee \neg q$
T	T	T	F	F	F	F
T	F	F	T	F	T	T
F	T	F	T	T	F	T
F	F	F	T	T	T	T

Hence, $\neg(p \wedge q) \equiv \neg p \vee \neg q$.

Example 4. Show that $p \Rightarrow q \equiv \neg p \vee q$.

p	q	$p \Rightarrow q$	$\neg p$	$\neg p \vee q$
T	T	T	F	T
T	F	F	F	F
F	T	T	T	T
F	F	T	T	T

The columns for $p \Rightarrow q$ and $\neg p \vee q$ are identical, so the equivalence holds.

Example 5. Prove $p \vee (p \wedge q) \equiv p$ (Absorption Law).

p	q	$p \wedge q$	$p \vee (p \wedge q)$	p
T	T	T	T	T
T	F	F	T	T
F	T	F	F	F
F	F	F	F	F

Both expressions give the same truth values; therefore, $p \vee (p \wedge q) \equiv p$.

(B) Validity of Arguments (Tautology Test)

Example 1.

- Premise 1: $p \Rightarrow q$
- Premise 2: p
- Conclusion: q (Modus Ponens)

p	q	$p \Rightarrow q$	p	$(p \Rightarrow q) \wedge p$	$((p \Rightarrow q) \wedge p) \Rightarrow q$
T	T	T	T	T	T
T	F	F	T	F	T
F	T	T	F	F	T
F	F	T	F	F	T

Since the final column is all T, the argument is **valid**.

Example 2.

- Premise 1: $\neg q$
- Premise 2: $p \Rightarrow q$
- Conclusion: $\neg p$ (Modus Tollens)

p	q	$p \Rightarrow q$	$\neg q$	$(p \Rightarrow q) \wedge \neg q$	$((p \Rightarrow q) \wedge \neg q) \Rightarrow \neg p$
T	T	T	F	F	T
T	F	F	T	F	T
F	T	T	F	F	T
F	F	T	T	T	T

Hence, the argument is **valid**.

Example 3.

- Premise 1: $p \vee q$
- Premise 2: $\neg p$
- Conclusion: q (Disjunctive Syllogism)

p	q	$p \vee q$	$\neg p$	$(p \vee q) \wedge \neg p$	$((p \vee q) \wedge \neg p) \Rightarrow q$
T	T	T	F	F	T
T	F	T	F	F	T
F	T	T	T	T	T
F	F	F	T	F	T

Thus, the argument is **valid**.

Example 4.

- Premise 1: $p \Rightarrow q$
- Premise 2: $q \Rightarrow r$
- Conclusion: $p \Rightarrow r$ (Hypothetical Syllogism)

p	q	r	$p \Rightarrow q$	$q \Rightarrow r$	$(p \Rightarrow q) \wedge (q \Rightarrow r)$	$((p \Rightarrow q) \wedge (q \Rightarrow r)) \Rightarrow (p \Rightarrow r)$
T	T	T	T	T	T	T
T	T	F	T	F	F	T
T	F	T	F	T	F	T
T	F	F	F	T	F	T
F	T	T	T	T	T	T
F	T	F	T	F	F	T
F	F	T	T	T	T	T
F	F	F	T	T	T	T

Therefore, the argument is **valid**.

Example 5.

- Premise 1: $p \vee q$

- Premise 2: $\neg q$
- Conclusion: p

p	q	$p \vee q$	$\neg q$	$(p \vee q) \wedge \neg q$	$((p \vee q) \wedge \neg q) \Rightarrow p$
T	T	T	F	F	T
T	F	T	T	T	T
F	T	T	F	F	T
F	F	F	T	F	T

Since all entries in the last column are T, this argument is **valid**.

Examples of Invalid Arguments

Example 6.

- Premise 1: $p \Rightarrow q$
- Premise 2: q
- Conclusion: p

p	q	$p \Rightarrow q$	$(p \Rightarrow q) \wedge q$	$((p \Rightarrow q) \wedge q) \Rightarrow p$
T	T	T	T	T
T	F	F	F	T
F	T	T	T	F
F	F	T	F	T

One F appears in the final column, so the argument is **invalid**.

Example 7.

- Premise 1: p
- Premise 2: q
- Conclusion: $p \Rightarrow q$

p	q	$p \wedge q$	$p \Rightarrow q$	$(p \wedge q) \Rightarrow (p \Rightarrow q)$
T	T	T	T	T
T	F	F	F	T
F	T	F	T	T
F	F	F	T	T

The implication does not always follow logically from the premises, hence **invalid**.

Example 8.

- Premise 1: $p \vee q$
- Premise 2: r
- Conclusion: p

p	q	r	$p \vee q$	$(p \vee q) \wedge r$	$((p \vee q) \wedge r) \Rightarrow p$
T	T	T	T	T	T
T	F	T	T	T	T
F	T	T	T	T	F
F	F	F	F	F	T

Because one case gives F in the last column, this argument is **invalid**.

Example 9.

- Premise 1: $p \Rightarrow q$
- Premise 2: r
- Conclusion: $p \Rightarrow r$

p	q	r	$p \Rightarrow q$	$(p \Rightarrow q) \wedge r$	$((p \Rightarrow q) \wedge r) \Rightarrow (p \Rightarrow r)$
T	T	T	T	T	T
T	F	F	F	F	T
F	T	T	T	T	T
F	F	T	T	T	T
T	T	F	T	F	T

Since all entries in the last column are T, this argument is **valid**.

Example 10.

- Premise 1: $\neg p$
- Premise 2: $q \Rightarrow p$
- Conclusion: $\neg q$

p	q	$\neg p$	$q \Rightarrow p$	$(\neg p) \wedge (q \Rightarrow p)$	$((\neg p) \wedge (q \Rightarrow p)) \Rightarrow \neg q$
T	T	F	T	F	T
T	F	F	T	F	T
F	T	T	F	F	F
F	F	T	T	T	T

As one row yields F, the argument is **invalid**.

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12 More Problems

1. Define a proposition. Give two examples of propositions and two that are not.
2. State and verify any one commutative law using a truth table.
3. What is a tautology? Give an example.
4. Simplify $\neg(p \to q)$.
5. Use truth tables to show that $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$.
6. Show that $\neg(p \to q) \equiv p \wedge \neg q$.
7. Distinguish between informal and formal reasoning.

8. Write the converse, inverse, and contrapositive of "If a number is divisible by 4, it is even."
9. Express "All humans are mortal" using quantifiers.
10. Verify using truth tables: (a) $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$, (b) $\neg(p \vee q) \equiv (\neg p \wedge \neg q)$.
11. Test the validity of: "If it rains, the ground is wet. It is raining. Therefore, the ground is wet."
12. Define predicate and quantifiers. Express in logic:
    (a) Every student likes mathematics.
    (b) Some students dislike statistics.
    (c) There exists a number that is both even and prime.
13. Explain any two theorem proving methods with examples.
14. Prove by contradiction that $\sqrt{2}$ is irrational.
15. Prove using contrapositive method that if n^2 is even, then n is even.

13 Answers

Answer Key

1. **Define a proposition. Give two examples of propositions and two that are not.**

   Answer: A *proposition* (or logical statement) is a declarative sentence that is either true or false (but not both).

   - Examples of propositions:
     - "$2 + 3 = 5$." (True)
     - "The Earth revolves around the Sun." (True)
   - Non-examples (not propositions):
     - "What time is it?" (A question — not true/false.)
     - "Let x be a number." (Open sentence / contains a free variable — not a proposition until x is specified.)

2. **State and verify any one commutative law using a truth table.**

   Answer: One commutative law: $p \wedge q \equiv q \wedge p$. We verify with a truth table:

p	q	$p \wedge q$	$q \wedge p$
T	T	T	T
T	F	F	F
F	T	F	F
F	F	F	F

   Each row shows identical values in the two columns, so $p \wedge q$ and $q \wedge p$ have the same truth table and are therefore equivalent.

   (You may similarly verify $p \vee q \equiv q \vee p$.)

3. **What is a tautology? Give an example.**

   Answer: A *tautology* is a compound proposition that is true under every possible assignment of truth values to its atomic propositions. Example: $p \vee \neg p$ (the law of excluded middle). Its truth table:

p	$\neg p$	$p \vee \neg p$
T	F	T
F	T	T

   The final column is always T, so the formula is a tautology.

4. **Simplify $\neg(p \to q)$.**

   Answer: Using the standard equivalence $p \to q \equiv \neg p \vee q$,

$$\neg(p \to q) \equiv \neg(\neg p \vee q).$$

   Apply De Morgan's law:

$$\neg(\neg p \vee q) \equiv \neg(\neg p) \wedge \neg q \equiv p \wedge \neg q.$$

   So $\neg(p \to q) \equiv p \wedge \neg q$.

5. **Use truth tables to show that $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$.**

   Answer: Construct the full truth table (8 rows for p, q, r):

p	q	r	$q \wedge r$	$p \vee (q \wedge r)$	$(p \vee q) \wedge (p \vee r)$
T	T	T	T	T	T
T	T	F	F	T	T
T	F	T	F	T	T
T	F	F	F	T	T
F	T	T	T	T	T
F	T	F	F	F	F
F	F	T	F	F	F
F	F	F	F	F	F

   Comparing the two final expression columns row by row we see they match in every row; hence

$$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r).$$

6. **Show that $\neg(p \to q) \equiv p \wedge \neg q$.**

   Answer: This is the same simplification as item 4. We can also verify with a truth table:

p	q	$p \to q$	$\neg(p \to q)$	$p \wedge \neg q$
T	T	T	F	F
T	F	F	T	T
F	T	T	F	F
F	F	T	F	F

The last two columns are identical, so $\neg(p \to q) \equiv p \wedge \neg q$.

7. **Distinguish between informal and formal reasoning.**

   Answer:

   - **Informal (natural language) reasoning:** Arguments expressed in ordinary language, relying on context, intuition, and background knowledge. Natural-language arguments are flexible and easy to communicate but can be imprecise, ambiguous, or contain hidden assumptions.
   - **Formal logical reasoning:** Uses precise symbolic language (e.g. $\wedge, \vee, \neg, \to, \forall, \exists$) and explicit rules of inference. Each step is justified; conclusions follow necessarily from premises. Formal reasoning reduces ambiguity and enables mechanical verification (truth-tables, proofs).

8. **Write the converse, inverse, and contrapositive of "If a number is divisible by 4, it is even."**

   Answer: Let p: "Number divisible by 4", q: "Number is even". Original implication: $p \to q$.

   - **Converse:** $q \to p$. (If a number is even, then it is divisible by 4.) — This is *not* generally true.
   - **Inverse:** $\neg p \to \neg q$. (If a number is not divisible by 4, then it is not even.) — This is *not* generally true.
   - **Contrapositive:** $\neg q \to \neg p$. (If a number is not even, then it is not divisible by 4.) — This is *logically equivalent* to the original statement and is true.

9. **Express "All humans are mortal" using quantifiers.**

   Answer: Let the domain be all persons; let $H(x)$ mean "x is human", and $M(x)$ mean "x is mortal". The statement is:

   $$\forall x\,(H(x) \to M(x)).$$

   If the domain is implicitly humans, one may shorten to $\forall x\, M(x)$ (but standard formalization uses the implication as above).

10. **Verify using truth tables:** (a) $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$, (b) $\neg(p \vee q) \equiv (\neg p \wedge \neg q)$.

    Answer: (a) This is identical to question (5); see the 8-row table there which shows equality of columns.

(b) De Morgan's law. Truth table:

p	q	$p \vee q$	$\neg(p \vee q)$	$\neg p$	$\neg q$
T	T	T	F	F	F
T	F	T	F	F	T
F	T	T	F	T	F
F	F	F	T	T	T

Now $\neg p \wedge \neg q$ column:

p	q	$\neg p \wedge \neg q$
T	T	F
T	F	F
F	T	F
F	F	T

Comparing $\neg(p \vee q)$ and $\neg p \wedge \neg q$ rows, they match exactly; hence the equivalence holds.

11. **Test the validity of: "If it rains, the ground is wet. It is raining. Therefore, the ground is wet."**

Answer: Let p: "It rains", q: "Ground is wet". Premises: $p \to q$, p. Conclusion: q. Form the implication

$$((p \to q) \wedge p) \to q.$$

Truth table:

p	q	$p \to q$	p	$(p \to q) \wedge p$	$((p \to q) \wedge p) \to q$
T	T	T	T	T	T
T	F	F	T	F	T
F	T	T	F	F	T
F	F	T	F	F	T

The final column is all T, so the formula is a tautology and the argument is **valid** (this is Modus Ponens).

12. **Define predicate and quantifiers. Express in logic:**

(a) Every student likes mathematics.

(b) Some students dislike statistics.

(c) There exists a number that is both even and prime.

Answer: A *predicate* is a formula $P(x)$ with a variable x that becomes a proposition when x is given a specific value. The *quantifiers* are: $\forall$ (universal) meaning "for all", and $\exists$ (existential) meaning "there exists".

Formalizations (choose domain = "students" where appropriate):

(a) Let $S(x)$ mean "x is a student", $L(x)$ mean "x likes mathematics". If domain is all people:

$$\forall x\,(S(x) \to L(x)).$$

If domain is "students," simply $\forall x\,L(x)$.

(b) Some students dislike statistics. Let $D(x)$ = "x dislikes statistics":

$$\exists x\,(S(x) \wedge D(x)).$$

If domain is students, $\exists x\,D(x)$.

(c) There exists a number that is both even and prime. Let domain be integers, $E(n)$ = "n is even", $P(n)$ = "n is prime":

$$\exists n\,(E(n) \wedge P(n)).$$

(This is true: $n = 2$.)

13. **Explain any two theorem proving methods with examples.**

Answer: We explain *Direct Proof* and *Proof by Contradiction*, each with a simple example.

(i) Direct Proof: To prove an implication $p \to q$, assume p and, using definitions and logical steps, derive q.

Example: Prove: If n is even, then n^2 is even.

Proof: Let n be even, so $n = 2k$ for some integer k. Then $n^2 = (2k)^2 = 4k^2 = 2(2k^2)$, which is even. □

(ii) Proof by Contradiction: To prove a statement S, assume $\neg S$ and deduce a contradiction (a statement of the form A and $\neg A$). Conclude that S must be true.

Example: Prove $\sqrt{2}$ is irrational. (Full proof given in Q14 below.) Assume $\sqrt{2}$ is rational, write $\sqrt{2} = a/b$ in lowest terms; derive both a and b even — contradiction. Hence $\sqrt{2}$ is irrational.

(Other methods: Contrapositive, Proof by Induction, Biconditional proofs etc.)

14. **Prove by contradiction that $\sqrt{2}$ is irrational.**

Answer: (Standard contradiction proof)

Assume to the contrary that $\sqrt{2}$ is rational. Then there exist integers a, b (with $b \neq 0$) such that

$$\sqrt{2} = \frac{a}{b},$$

and assume this fraction is in lowest terms ($\gcd(a, b) = 1$). Square both sides:

$$2 = \frac{a^2}{b^2} \quad \Rightarrow \quad a^2 = 2b^2.$$

So a^2 is even, which implies a is even. Thus $a = 2k$ for some integer k. Substitute:

$$(2k)^2 = 2b^2 \quad \Rightarrow \quad 4k^2 = 2b^2 \quad \Rightarrow \quad b^2 = 2k^2.$$

Hence b^2 is even, so b is even. Thus both a and b are even, contradicting the assumption that a/b was in lowest terms. Therefore, $\sqrt{2}$ is irrational. $\square$

15. **Prove using contrapositive method that if n^2 is even, then n is even.**

Answer: We prove the contrapositive: if n is odd, then n^2 is odd. The contrapositive is logically equivalent to the original statement.

Assume n is odd, so $n = 2k + 1$ for some integer k. Then

$$n^2 = (2k + 1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1,$$

which is of the form $2m + 1$, an odd integer. Hence if n is odd then n^2 is odd. Therefore, the contrapositive of the original statement is true, and so the original statement holds: if n^2 is even then n is even. $\square$